

ColorVision

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Overview

ColorVision app is designed to provide users with an accessible and interactive platform to test and manage their color vision capabilities. It includes various color vision tests, user profile management, customizable UI settings (e.g., dark mode, high contrast), and data persistence features. This app aims to empower individuals with color vision deficiencies by providing insights into their condition through a simple, user-friendly interface.

The following requirements are covered in our app:

1. Navigation and Event-Based Action

Covered:

- **Navigation:** The app uses a navigation structure that allows users to move between various screens, such as:
 - **Sign-In** and **Sign-Up** screens for authentication.
 - **Home Screen** after login, where users can access different tests.
 - **Test Screens** (e.g., Ishihara Test) accessed from the Home Screen.
 - **Edit Profile Screen** for managing user details.
- **Event-Based Actions:**
 - **Sign-In/Sign-Up Actions:** When a user enters their credentials and clicks "Sign In" or "Sign Up," the app verifies/stores the data and navigates to the Home Screen.
 - **Toggle Actions:** Users can toggle between **Dark Mode** and **High Contrast** settings. These actions trigger changes in UI color themes based on user preferences.
 - **Quiz Interaction:** Starting a quiz triggers a timer and presents questions in a timed, sequential manner, where user actions (selecting answers) affect the score and progress through the quiz.

Justification: Navigation and event-based actions allow for a smooth user experience by enabling easy access to all key functions (authentication, tests, settings) and providing interactive feedback based on user input.

2. Storing/Retrieving Data

Covered:

- **User Authentication Data:** User credentials (e.g., email and password) are stored locally upon signup, allowing the app to retrieve and verify login information on subsequent sign-ins.
- **User Preferences:** Settings like **Dark Mode** and **High Contrast** preferences are saved using AsyncStorage, enabling the app to retrieve and apply these settings each time the user logs in.
- **Quiz Data:** The app retrieves quiz content (e.g., images, options) when the user starts a test and stores the user's responses temporarily for scoring.

Justification: Storing and retrieving data locally ensures a personalized experience for users by preserving their settings and authentication details, making the app more user-friendly and responsive.

3. Basic UI/UX to Demonstrate Expected Behavior

Covered:

- **Sign-In/Sign-Up Screens:** Clear input fields for credentials and a user-friendly layout guide users through the login process.
- **Home Screen with Test Options:** The Home Screen displays tiles with different test names, visually organizing the app's main functions. This screen includes a profile icon for easy access to the profile page, a dark mode toggle, and a logout button.
- **Test Screen (Quiz):** The quiz screen features images, answer options, and a timer to provide an interactive and engaging experience.
- **Edit Profile Screen:** Allows users to view and update their profile information, maintaining a consistent and intuitive experience.
- **Styling:** The app uses color themes that switch based on Dark Mode and High Contrast settings, improving accessibility and visual appeal.

Justification: The UI/UX elements ensure the app is intuitive, visually organized, and accessible. The interface demonstrates the app's core functions clearly, allowing users to navigate and interact with ease.

The following outlines the key aspects of the app, organized as follows: (1) a list of necessary features for the Minimum Viable Product (MVP), (2) a breakdown of development tasks, and (3) an explanation of how each feature adds value to the user experience.

1. List of Features Needed for the Minimum Viable Product (MVP)

1. User Authentication

- **Feature:** Sign-up and sign-in functionality with local storage for user data.
- **Justification:** Essential for user identity management, enabling personalized experiences and data persistence.

2. User Profile Management

- **Feature:** Allow users to view and edit their profile information, such as age, gender, and password.
- **Justification:** Supports customization and allows users to maintain accurate information, improving user retention and personalization.

3. Color Vision Tests

- **Feature:** Include multiple color vision tests like Ishihara Test, Color Arrangement Test, Cambridge Color Test, and RGB Anomaloscope Test.
- **Justification:** Provides core functionality to assess color vision, which is the primary purpose of the app, adding direct value to users seeking color vision evaluation.

4. Quiz Functionality for Tests

- **Feature:** Start a timed quiz for each test, showing an image and multiple-choice options for each question.
- **Justification:** Engages users interactively, offering real-time feedback on their color vision capabilities.

5. Settings Management

- **Feature:** Toggle options for dark mode and high contrast, with preferences saved per user.
- **Justification:** Enhances accessibility by catering to users' visual comfort needs and ensuring settings persist across sessions, adding significant value for users with color vision deficiencies.

6. Logout Option

- **Feature:** Allow users to log out, clearing the session data but retaining their stored preferences.
- **Justification:** Necessary for security, allowing users to manage access to their accounts.

2. Breakdown of Tasks

Task	Description
Setup User Authentication	Implement sign-up and sign-in screens and save user data locally for persistence.
Build User Profile Screen	Create an interface to display and update user profile information like age, gender, and password.
Implement Color Vision Tests	Develop individual screens for each test (e.g., Ishihara Test) with image and question options.
Quiz Functionality for Tests	Develop quiz mechanics with a timer, multiple-choice options, and score calculation.
Settings Management (Dark/High Contrast)	Add toggle buttons for dark mode and high contrast mode and save settings in local storage.
Logout Implementation	Provide a logout option to clear session data and return to the sign-in screen.
Styling and Basic UI/UX Enhancements	Refine UI components for improved accessibility and visual appeal.
README Update	Update README.md to reflect project details, team members, and app functionalities.

3. Mapping Between Features and Value(s) Delivered by the App

Feature	Value to Users
User Authentication	Ensures a personalized experience by storing user-specific settings and quiz progress.
User Profile Management	Provides users with control over their information, allowing personalization based on age and gender.
Color Vision Tests	Directly addresses the primary purpose of the app by offering scientifically-backed color vision tests.
Quiz Functionality	Increases engagement by providing an interactive, time-bound quiz experience.
Dark Mode & High Contrast	Enhances accessibility for visually impaired users by offering customizable visual settings.
Logout Option	Ensures user security and privacy by allowing them to manage their session data.
Basic UI/UX	Provides an intuitive and visually appealing interface, making the app easy to navigate and enjoyable to use.